

SERGIO RODRIGUEZ VALBUENA

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PROFILE

AI Engineer with 2 years of experience developing Software and Machine Learning solutions. I hold a BSc in Data Science and Engineering from Universidad Carlos III de Madrid, where I was awarded an Excellence Scholarship and spent one year abroad at the University of California. I am proactive and passionate about creating innovative solutions that provide real value.

EDUCATION

- **Universidad Carlos III de Madrid** - Bachelor's degree in Data Science and Engineering Sept 2021 – Jun 2025, Madrid
GPA: 8/10 (Excellence Scholarship of Madrid 2025)
- **University of California, Riverside** - Computer Science (Exchange Program) Aug 2023 – May 2024, Los Ángeles
GPA: 3.6/4

EXPERIENCE

- **Plexus Tech** - AI Engineer Sept 2025 – Today, Madrid
 - Developed multimodal RAG systems, integrating different vector database providers, orchestrating retrieval and re-ranking flows and exposing RAG functionality as tools through MCP in containerized applications with Docker.
 - Applied research and development of ML and DL models for Network Intrusion Detection Systems, training and evaluating different architectures on network traffic data, with tracking through MLflow and data ingestion supported by Kafka.
 - Developed ML models for time-series forecasting and anomaly detection using industrial sensor data, applying Conformal Prediction to provide statistical guarantees for predictions.
 - Designed conversational AI systems, including a real-time voice agent prototype for customer service environments, integrating ASR and TTS models with LLM reasoning and guardrails.
 - Production deployment of a RAG-based classification system for aviation safety incident reports, built on hexagonal architecture, integrating AWS Bedrock and Neural Trust guardrails, and optimizing accuracy and cost.

- **MissioIA** - AI Software Engineer (Part-time) Oct 2024 – Jun 2025, Madrid
 - Developed software solutions in a fast-paced environment, working closely with clients to analyze their systems and business requirements, integrating generative AI to automate workflows and optimize their processes.
 - Built full-stack applications with Python and React (Next.js), following best practices such as TDD or DDD, version control, code quality, testing and CI/CD.
 - Applications were deployed on Google Cloud Platform and supported by Firebase for authentication and storage.

- **PRGX Global Inc.** - Data Analyst I (Internship) Sept 2024 – Mar 2025, Madrid
 - Automated ETL processes with Python and DBT.
 - Performed data analysis and reports using SQL and Microsoft SQL Server.

- **Hito 1** - Machine Learning Engineer (Internship) Jun 2024 – Aug 2024, Asturias
 - Performed EDA and feature engineering on hydrological time-series data to optimize a ML model for predicting spillway overflow events in Asturias.

PROJECTS

- **AI-Driven System for Automated Project Design** - Bachelor Thesis associated with MissioIA
 - Developed a multi-agent AI system capable of designing software project prototypes in GCP using MCPs.
 - Implemented specialized agents for backend, frontend, and deep research tasks, coordinating them through shared memory structures, synchronization points, and task delegation mechanisms.

- **Samplex – AI-Powered Study Assistant** - Course Project · Universidad Carlos III de Madrid
 - Developed an Android app that generates personalized study plans from uploaded materials.

- **Mobility and Routine Analysis for Mental Health Assessment** – Course Project with eB2 (Evidence-Based Behaviour)
 - Analyzed large-scale geolocation data from mobile devices to uncover users' mobility patterns and daily routines.
 - Applied clustering algorithms (DBSCAN), entropy analysis, and time-series analysis to extract behavioral insights.

- **Deep Learning model for Waste Classification** - Course Project · University of California
 - Developed a waste recognition system using CNNs to classify discarded items and recommend appropriate recycling bins.
 - Achieved 90.3% accuracy evaluating different open-source architectures (EfficientNetB0, ResNet50, MobileNet, DenseNet).

CERTIFICATIONS

- **Azure AI Engineer Associate** – Microsoft, Jun 2026.
- **Claude Certified Architect** – Foundations (CCA-F) – Anthropic, Apr 2026.
- **C1 Advanced** (CEFR C1 English Certificate) – University of Cambridge, Dec 2021.